

Ruize Li

📍 Jiangsu, China ✉ ruizeli@smail.nju.edu.cn ☎ +86 139 0917 6526 📧 lixrui-del

Education

Nanjing University & University of Helsinki
B.Sc. Dual Degree Program in Atmospheric Sciences

Nanjing, China
Sept 2023 – Present

- **GPA:** 4.48/5.00 (Rank: 19/77)
- **Honors:** **Excellent Student of Nanjing University (Top ~5%, 2024)**, People's Scholarship (Second Prize, 2024), **Meritorious Winner** in MCM (**Top ~7%, 2026**)
- **Core Coursework:** Dynamic Meteorology, Principles of Synoptic Meteorology, Aerosol Physics, Probability & Statistics, Linear Algebra
- **CS & AI Electives:** Deep Learning Platforms & Applications, Introduction to Machine Learning, Data Mining, Data Structures & Algorithms, Introduction to Optimization

Hong Kong University of Science and Technology (HKUST)
Fully Funded Exchange Student / School of Engineering

Hong Kong
Feb 2026 – June 2026

- Conducted **AI for Science** research in Prof. Song Guo's group and completed the RealBench paper and open-source benchmark as a co-first author.
- **Core Coursework:** Large-Scale Machine Learning System for Foundation Models, Discrete Math.

Publications

“RealBench: Benchmarking Data-Driven Numerical Weather Forecasting Under Operational Conditions and Extreme Event Challenges”

Under Review

Ruize Li*, Zhibin Wen*, Tao Han*, Hao Chen, Fenghua Ling, Wei Zhang, Song Guo, Lei Bai

(**NeurIPS 2026 Under Review, Evaluations and Datasets Track**) [\[arXiv\]](#) [\[Code\]](#)

* denotes equal contribution by three first authors

Projects & Research Experience

Comprehensive Benchmark for AI-based Medium-Range Weather Forecasting

Hong Kong, China
Nov 2025 – Present

- Developing a standardized evaluation framework to compare SOTA AI models (**Pangu-Weather**, **GraphCast**, **Aurora**) against operational **ECMWF IFS** forecasts under **real-world conditions**.
- Incorporating real-time operational data and station observations as ground truth to address the inherent 4D-Var assimilation biases of idealized **ERA5** reanalysis.
- Conducting inference and evaluation on GPU clusters, providing a robust baseline to quantify performance on **rare, high-impact extreme events** (e.g., tropical cyclones).

Mathematical Contest in Modeling (MCM) 2026
Team Leader & Student Advisor

Nanjing, China
Feb 2026

- **Awarded Meritorious Winner (First Prize, Top ~7% Globally)** for tackling Problem A (Continuous Physical Modeling).
- Developed a closed-loop dynamical system modeling framework, coupling physical state evolution (**ODEs**) with external empirical drivers, solved via the **RK45** numerical method.
- Implemented an **Extended Kalman Filter (EKF)** for **data assimilation**, effectively fusing noisy observational telemetry to correct trajectory drift and achieve robust state estimation ($R^2 = 0.936$).
- Conducted comprehensive **Uncertainty Quantification (UQ)** via **Monte Carlo** simulations and parameter **sensitivity analysis**, assessing model robustness under extreme environmental perturbations.

Optimizing Contextual Lipschitz Bandits via Local Data Aggregation

Research Intern / Advised by Prof. Lin Yang

Nanjing, China

Sept 2024 – Jan 2025

- Designed an advanced **Multi-Armed Bandit (MAB)** algorithm to optimize decision-making in continuous state spaces.
- Proposed a novel **‘Local Data Combining’** mechanism leveraging Lipschitz continuity to accelerate the pruning of sub-optimal arms by sharing observations across similar contexts.
- Provided theoretical analysis on the regret bound of the algorithm, demonstrating improved convergence rates.

Reproducing VQ-Prompt under the LibContinual Framework

Course Project / Deep Learning & Continual Learning

Nanjing, China

June 2025 – Aug 2025

- Fine-tuned a **Vision Transformer (ViT)** backbone and transferred the Prompt module to adapt the model architecture to the LibContinual framework.
- Identified architectural discrepancies between CODA-Prompt and VQ-Prompt and modified the attention mechanism for adaptation.
- Achieved reproduction accuracy within **2-4%** of the original baseline, validating the effectiveness of the prompt-based continual learning approach.

Future Changes in East Asian Summer Rainfall: PDO & Global Warming

Research Student / Advised by Prof. Sandro M. Ferreira Veiga

Helsinki, Finland

June 2025 – Sept 2025

- Researched future changes in East Asian Summer Rainfall driven by PDO and global warming.
- Implemented **Empirical Orthogonal Function (EOF)** analysis to extract dominant PDO signals from complex North Pacific SST anomaly fields.
- Quantified and validated that PDO-associated precipitation patterns remain robust under future climate scenarios, providing insights for long-term climate risk assessment.

Skills

Programming: Python (Advanced), C++

AI & Data Stack: PyTorch, Scikit-Learn, NumPy, Pandas, Xarray, Dask, Cartopy

Tools & Environment: Git, Linux/Shell, Slurm, tmux, Docker, LaTeX, VS Code

Language: English (CET-6: 553)

Extracurricular Activities

- **Director**, Academic Innovation Department, Student Union, NJU Nanjing-Helsinki Institute (2025–2026): Led the *Atmospheric Science Major Survival Handbook* and *Feiyue Handbook*, designing their outlines, writing core content, and coordinating contributions from students and alumni.
- Built and maintained the “Gongxue Community.”
- **Selected Participant**, 2026 National Forum for Outstanding Undergraduate Students in Geosciences, Department of Earth System Science, Tsinghua University: Delivered a RealBench research presentation and completed the selection interview.
- **President**, Shaanxi Provincial Association of Admissions Counselors: Organized recruitment campaigns and freshmen orientation for over 500 students.
- Cumulative volunteer service hours exceed **200 hours**.